Photovoltaic and Battery Storage Evaluation for Commercial Buildings in Sharjah

Israel Owolabi and Mariya Sokolovskaya
Skolkovo Institute of Science and Technology (Skoltech)
Moscow, Russia

***Abstract*—Battery storage increases photovoltaic (PV) self-consumption, but enclosure cooling and low utilization can limit its value. This paper evaluates 50 kWp of PV and a 50 kW/200 kWh energy storage system for three synthetic commercial buildings in Sharjah, United Arab Emirates. Annual 15-minute power balances are coupled to a container heat balance and a 20-year cash-flow model. Lithium iron phosphate (LFP) and vanadium redox flow batteries (VRFB) are compared against the same buildings with PV alone. When cooling, pumps and controls are counted, the total annual storage efficiency is 77–81% for insulated LFP and 61–64% for insulated VRFB. At an import price of 0.12 $/kWh and no payment for exports, the 20-year net present value is -78 to -64 k$ for LFP and -116 to -107 k$ for VRFB, so neither battery breaks even. Smaller storage, a better enclosure and additional revenue reduce the loss. A hypothetical evening export price of 0.60 $/kWh makes retail LFP profitable with ideal forecasts; VRFB would need about 0.97 $/kWh. PV alone pays back, so PV and storage should be decided separately and storage sized to defined services.**

***Index Terms*—photovoltaics, battery energy storage, container cooling, LFP, vanadium redox flow battery, self-consumption, net present value**

I. Introduction

The United Arab Emirates (UAE) has a strong solar resource, but photovoltaics (PV) produce power at midday while commercial buildings also use electricity in the evening and at night. A battery can shift surplus PV, lower the peak grid import or carry loads through an outage. These services need different sizing and have different value, so high self-consumption alone does not make a battery a good investment.

The case studied is a commercial building in Sharjah with 50 kWp of PV and a 50 kW/200 kWh energy storage system (ESS), four hours at full power, in an outdoor container designed for 45 °C ambient air. The study asks how much PV energy reaches useful loads once the storage system's own electricity use (cooling, pumps, controls) is counted, and when the battery pays for itself. Lithium iron phosphate (LFP) and vanadium redox flow batteries (VRFB) are compared for office, retail and 24/7 buildings such as hospitals, each using 100 MWh a year.

A. Findings from Previous Studies

Table I summarizes the main findings of the literature review. The container studies start from cool supply air, so at 45 °C the energy needed to make that air must be computed. LFP aging links the container temperature to the energy delivered, and storage rarely pays under flat energy tariffs.

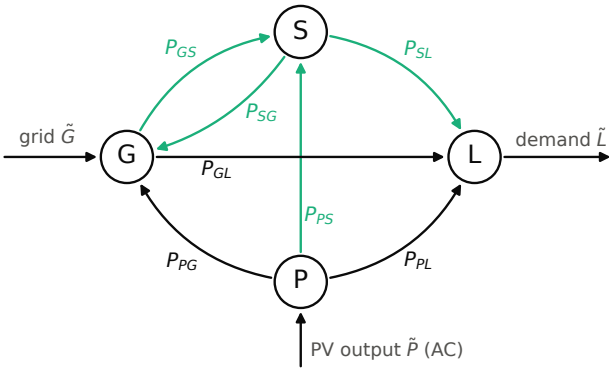


Fig. 1. Power-flow network of (2)–(5). Green edges pass through the PCS and carry the one-way efficiency η.

The review did not find a study that links envelope heat gain, cooling energy, delivered energy, aging and cost for a containerized battery in a hot climate. Olis *et al.* link the envelope, cooling and battery size for cold sites [4], Yagci *et al.* give LFP aging at cell level [5] and Alkaabi *et al.* give UAE economics without a thermal model [10]. This paper makes that link for one site. It 1) estimates the design-day heat load and the energy the battery can deliver, 2) simulates a year of 15-minute power flows with container cooling in the loop for both chemistries and 3) evaluates the cash flows and the conditions under which each battery breaks even. It is a scenario study, not a monitored installation or a detailed electrical or safety design.

II. Materials and Methods

A. System Boundary and Synthetic Inputs

Fig. 1 shows the model, a directed network of PV (P), storage (S), load (L) and grid (G) nodes. PV enters at its AC output, and the four storage edges pass through the power conversion system (PCS); a shared DC bus and converter part-load behavior are not modeled.

The model runs one year in 15-minute steps ($\Delta t = 0.25$ h). Ambient temperature and PV output are reconstructed from monthly climate data with standard temperature-profile and solar-geometry methods [1], [15]. Monthly sunshine hours give the daily solar energy on a horizontal surface by the Angstrom–Prescott relation, with the default coefficients 0.25 and 0.50 of [16]. This energy is spread over the day and

TABLE I
FINDINGS OF THE LITERATURE REVIEW THAT SET THE DESIGN BASIS

Finding	Use in this work
Climate. The cooling dry-bulb temperature exceeded in 0.4% of the year at Sharjah airport is 44.2 °C [1].	45 °C is a design-day peak
Container cooling. CFD studies supply 15–20 °C air and omit the envelope; the converter is the hot spot [2], [3]. Cooling and heating used 8–34% of the charged energy; ignoring it made batteries 42–300% too small (cold sites, COP 3) [4].	Envelope varied; own use added to the building load
LFP aging. LFP cells at 0.28C lost 11% at 35 °C and 22.3% at 50 °C in 1500 cycles; activation energy 37.3 kJ/mol [5].	Air held at 30 °C; fade estimate
Chemistry. LFP: 80–160 Wh/kg, 90–94% round trip, 3000–10 000 cycles. VRFB: 15–70 Wh/kg, 65–85%, 10 000–20 000 cycles. Lead-acid fails on cycle life and space [6].	LFP and VRFB compared
VRFB in heat. No V_2O_5 precipitation at 50 °C, even near full charge [7]; 200 kWh needs 8.3 m^3 of electrolyte [8].	Feasible, but bulky
Safety. 57% of reported failures came in the first two years; 18% of audited systems had thermal defects [9].	Fire safety still needed
Economics. Abu Dhabi: PV alone 0.058 \$/kWh vs a 0.054 \$/kWh tariff, with storage 0.17–0.21 \$/kWh [10]. PV paid back in 84% of scenarios, PV with a battery in 0.9% (reported in [11]); under a flat rate the cost-optimal design had no battery [12].	Battery likely needs value beyond the bill
Rules. DEWA's code covers PV only [13]; SEWA bills energy in slabs, with no demand or time-of-use charges [14].	Only self-consumption pays

converted to the sunlight on south-facing modules tilted at 23°, including reflected light (albedo 0.25). The cell temperature is

$$T_c = T_a + \frac{\mathrm{NOCT} - 20}{800} G_T, \tag{1}$$

where G_T is the sunlight on the module surface in W/m^2, T_a the air temperature and NOCT = 45 °C the cell temperature at 800 W/m^2 and 20 °C air. PV power falls by 0.35% per kelvin of cell temperature above 25 °C. One 14% allowance covers inverter, wiring, module mismatch and soiling between regular cleanings, and output is capped at the 50 kW inverter rating.

The office is occupied Monday to Friday, 08:00–18:00, the retail building every day, 10:00–22:00, and the third building never closes (24/7). Air conditioning is 60% of annual demand, typical of UAE buildings [17]; the rest follows occupancy and a base load. Each profile is scaled to 100 MWh a year, so only the timing differs (Fig. 2).

B. Power-Flow Constraints and Operating Rules

Each flow $P_{ij} \geq 0$ (kW) is measured where it leaves node i. In every step t the PV, load and grid nodes balance, and storage follows its energy balance:

$$\tilde{P} = P_{PL} + P_{PS} + P_{PG}, \tag{2}$$

$$\tilde{L} + P_a = P_{PL} + \eta P_{SL} + P_{GL}, \tag{3}$$

$$\tilde{G} = P_{GL} + P_{GS} - P_{PG} - \eta P_{SG}, \tag{4}$$

$$E_{t+1} = \sigma E_t + \Delta t\,[\eta(P_{PS} + P_{GS}) - P_{SL} - P_{SG}], \tag{5}$$

where $\tilde{P}$ is the PV AC output, $\tilde{L}$ the building demand, P_a the storage system's own use (Section II-C; zero in the runs without auxiliaries), $\tilde{G}$ the net grid import (negative when PV is exported) and E the stored energy. Only the storage edges lose energy: $\eta = \sqrt{\eta_{\mathrm{DC,RT}}}\,\eta_{\mathrm{PCS}}$ is their one-way efficiency, and σ is an assumed self-discharge of 2% a month for both technologies. The physical limits are

$$\tilde{P},\ P_{PS},\ P_{GS},\ P_{SL},\ P_{SG} \leq 50 \text{ kW}, \tag{6}$$

$$E_{min} \leq E_t \leq E_{max}, \tag{7}$$

$$(P_{PS} + P_{GS})(P_{SL} + P_{SG}) = 0. \tag{8}$$

TABLE II
STORAGE AND AUXILIARY INPUTS AT 200 KWH NAMEPLATE

Parameter	LFP	VRFB
Power rating (kW)	50	50
Usable energy (kWh)	180	200
DC round trip (%)	96	83
PCS efficiency, each way (%)	97.5	97.5
Nominal AC round trip (%)	91.3	78.9
Pump power, share of throughput (%)	0	3
Controls and BMS, continuous (kW)	0.30	0.30

Efficiencies from [6], [18]; pumps and controls are assumed values.

The 50 kW limits are the inverter and converter ratings; the grid edges have no physical limit, so the grid is the slack. LFP works between 5 and 95% state of charge (SOC), so $E_{min} = 10$ and $E_{max} = 190$ kWh; the VRFB uses its full 200 kWh. The last condition forbids charging and discharging in the same step. In every run the node balances hold to machine precision. Table II gives the storage inputs.

Three operating rules are tested without storage auxiliaries. Rule A maximizes self-consumption: surplus PV charges the battery, which discharges before the grid supplies. Rule B discharges only at 06:00–09:00 and 16:00–22:00, after a 10 kW night top-up to 50% SOC. Rule C keeps 60 kWh in reserve to hold grid import below the lowest cap that holds all year. The coupled thermal and economic cases use Rule A.

C. Container Heat Balance and Own Use

Fig. 3 shows the heat and electricity flows. The container measures $6.06 \times 2.44 \times 2.59$ m and its air is held at 30 °C. Roof and wall gains use the sol-air method [1]:

$$T_{sa,i} = T_a + \alpha I_i / h_o - \Delta T_{sky,i}, \tag{9}$$

$$Q_e = \textstyle\sum_i U_i A_i (T_{sa,i} - T_{in}) + Q_{floor}, \tag{10}$$

$$U_i = (h_o^{-1} + R_{ins,i} + h_i^{-1})^{-1}, \tag{11}$$

with irradiance I_i on each surface, film coefficients $h_o = 22.7$ and $h_i = 8.3$ W/(m^{2}K), a roof sky correction of 4 K and a floor conductance of 0.8 W/(m^{2}K). Four envelopes are tested:

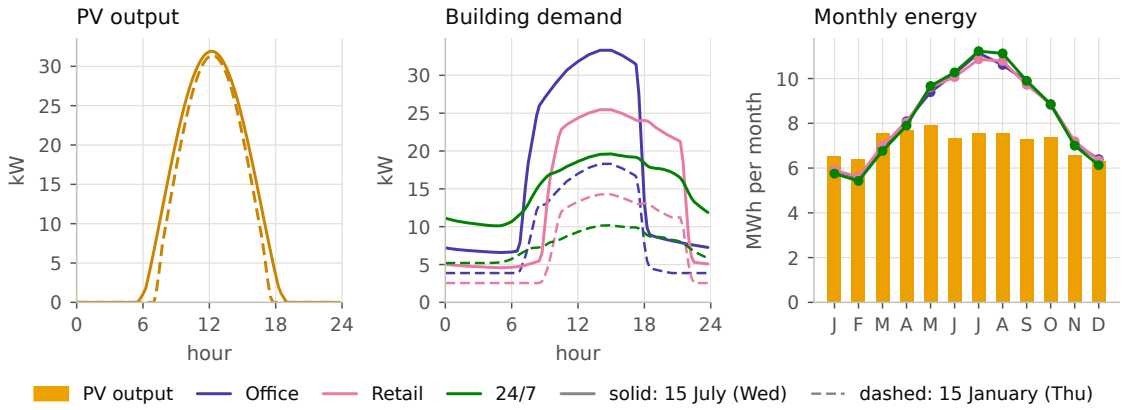


Fig. 2. Synthetic PV output and building demand on 15 July (solid) and 15 January (dashed), and monthly energy. All three buildings use 100 MWh a year.

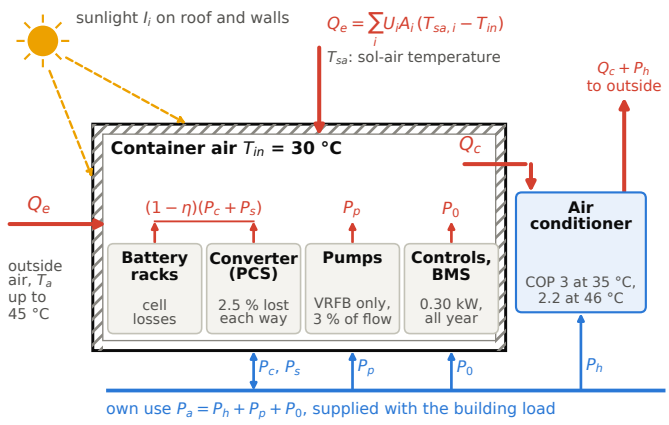


Fig. 3. Heat (red) and electricity (blue) flows of the containerized ESS. The hatched band is the insulation behind the steel skin. Schematic, not to scale.

bare dark steel ($\alpha = 0.90$); white coating ($\alpha = 0.35$); white coating with 50 mm polyurethane (PU), $R_{ins} = 2$ m^{2}K/W, the reference case; and 100 mm PU, $R_{ins} = 4$ m^{2}K/W, with a shaded roof (roof sun $\times 0.21$).

Design day. A steady estimate at $T_a = 45\,^\circ$C, with peak sun on the roof (950 W/m^2) and on all four walls at once (400 W/m^2) and rated power, gives the peak cooling load. Internal heat is

$$Q_{int} = P(1 - \sqrt{\eta_{\mathrm{DC,RT}}}) + P(1 - \eta_{\mathrm{PCS}}) + P_0 = 2.56 \text{ kW}, \tag{12}$$

with $P = 50$ kW and controls $P_0 = 0.30$ kW. The energy the battery delivers in year 10 follows a chain from the 200 kWh label: a 90% usable window; LFP aging measured at 35 °C [5], scaled to 30 °C air by the Arrhenius factor $f_T = \exp[-(E_a/R)(1/T_{in} - 1/T_{test})] = 0.787$; calendar (8% in 850 days) and cycle (11% in 1500 cycles) losses extended to 10 years and 3650 cycles by a square-root law and added,

$$L_{10} = f_T\left[0.08\sqrt{3650/850} + 0.11\sqrt{3650/1500}\right], \tag{13}$$

which gives 26.5%; and the 8–34% cooling share of [4]. Adding both losses at one full cycle a day makes the chain cautious.

Annual model. Battery and converter losses heat the container. With total charge and discharge powers P_c and P_s at their source nodes, pump power P_p and controls P_0, the internal heat and the heat removed in each step are

$$Q_{int} = (1-\eta)(P_c + P_s) + P_p + P_0, \tag{14}$$

$$Q_c = \max(0,\, Q_e + Q_{int}), \tag{15}$$

$$P_a = Q_c/\mathrm{COP}(T_a) + P_p + P_0, \tag{16}$$

$$\mathrm{COP}(T_a) = \min\left[5,\; 3\,\frac{35 + 12 - 15}{T_a + 12 - 15}\right]. \tag{17}$$

The COP scales a rating of 3 at 35 °C [4] by the Carnot ratio with a 15 °C evaporator and a condenser 12 K above ambient, giving 2.23 at 46 °C, the hot-climate rating point of [19]. Heating and thermal mass are omitted. Own use P_a enters the load balance (3); dispatch and (16) are iterated until they agree within 10^{-9} kW. The annual total storage efficiency is

$$\eta_{tot} = \frac{E_{out} + \eta(E_{end} - E_{start})}{E_{in} + E_{HVAC} + E_p + E_{ctrl}}, \tag{18}$$

where the numerator corrects for the change in stored energy. Unlike a laboratory round trip, it counts all auxiliary energy and depends on how much the battery is used.

D. Economic Model

The battery is judged against the same building with the same PV and no battery, in real US dollars over 20 years at an 8% discount rate [20]. Every avoided kWh is valued at SEWA's top commercial slab including the fuel surcharge, 0.44 AED/kWh = 0.12 $/kWh [14]. Exports earn nothing and there is no demand charge in the base case; slab changes within the month and VAT are ignored. Table III gives the costs and what each installed price pays for.

The LFP price is that of a Li-ion system for buildings, 339 $/kWh, plus 25% for installation [20]. The VRFB price is the present-day cost of [8] at 2 and 10 hours, fitted as $c(h) = a + b/h$ and read at 4 hours (1.08 $/€). It is for MW-class plants and excludes site installation. LFP capacity falls linearly to 73.5% at year 10, and its yearly benefit is interpolated between runs at 100, 80 and 60% capacity.

TABLE III
ECONOMIC INPUTS FOR 4-HOUR SYSTEMS, 200 KWH AND 50 KW

	LFP	VRFB
Installed price ($/kWh)	424	591
battery pack incl. BMS	199	–
electrolyte and tanks	–	263
stacks	–	202
converter (PCS)	24	40
container, cooling, wiring	116	–
pumps, piping, wiring, controls	–	85
site installation	85	–[a]
Initial cost C_0 (k$)	84.8	118.1
O&M ($/kW per year)	10	28
Life; replacement	15 y; year 15, $0.5C_0$	20 y; none
Value left at year 20, S_{20}	$(10/15)\,0.5C_0$	0
Capacity at year 10	73.5%	100%[b]

Sources: LFP price [20], pack and converter [21], life [21], replacement [22]; VRFB price and parts [8]; O&M [18]; LFP fade from the design-day chain. [a] Not included. [b] Rebalancing is part of O&M.

With $d_y = (1+r)^{-y}$, the yearly benefit and the net present value (NPV) are

$$B_y = p_{imp}\Delta E_{imp,y} - p_{exp}\Delta E_{exp,y}, \tag{19}$$

$$\mathrm{NPV} = -C_0 + \textstyle\sum_{y=1}^{20}(B_y - M - R_y)d_y + S_{20}d_{20}, \tag{20}$$

where ΔE_{imp} is grid import avoided net of own use, ΔE_{exp} is PV export given up, and M, R_y and S_{20} are O&M, replacement and value left. Simple payback is $T = C_0/(B_1 - M)$. The levelized cost of storage (LCOS) is

$$K_y = M + R_y + p_{imp}E_{a,y} + p_{exp}E_{in,y},$$
$$\mathrm{LCOS} = \frac{C_0 + \sum_y K_y d_y - S_{20}d_{20}}{\sum_y E_{out,y}d_y}, \tag{21}$$

with own use E_a valued at the import price and PV charging at the export price given up. NPV is linear in the battery price c and the tariff, so the break-even price c^* and tariff p^* are exact. p^* is the discounted cost of each kWh of grid import avoided, not the LCOS, and $V = -\mathrm{NPV}/\sum_y d_y$ is the yearly value that other services, such as backup, would have to add.

The economic size sweep (50–300 kWh) keeps four hours, unlike the physical sweep at a fixed 50 kW. Scenario 2 gives the VRFB 70% of its energy-component cost back at year 20; a variant sells LFP for nothing and VRFB for 50% of C_0. Scenario 3 pays 0.30 or 0.60 $/kWh for exports at 18:00–22:00 and nothing at other times. It uses an hourly mixed-integer dispatch with ideal forecasts that charges from PV only, returns to minimum SOC each day and keeps the Scenario 2 value left. The PV-only reference gets the same export rights. To find where VRFB would be the better buy, its future cost of 391 $/kWh at 4 h [8], a 30-year life and resale of its electrolyte at 70% of its price are also tested.

III. RESULTS

A. Design-Day Heat Load and Delivered Energy

At the design point a bare steel container must remove 15.4 kW of heat, and 83% of it enters through the roof and walls. Inside, the converter releases more heat than the cells, because the battery works at only 0.25C. White coating alone lowers the load by about 30%; with 50 mm PU it falls almost five-fold, to 3.3 kW, and 100 mm PU with a shaded roof adds little. The envelope, not the battery, sets the size of the cooling plant.

The 200 kWh label gives 180 kWh usable when new, 132 kWh at year 10 and 87–122 kWh after cooling, so the label can overstate the energy delivered by up to 2.3 times. About 300 kWh would have to be installed to keep 200 kWh usable at year 10.

TABLE IV
OPERATING RULES WITHOUT STORAGE AUXILIARIES

	Office	Retail	24/7
Annual grid import (MWh)			
PV only	38.6	44.6	49.7
A: self-consumption	18.8	17.7	18.6
B: peak windows	26.8	21.1	26.7
C: grid cap	21.3	18.1	18.6
Highest grid import (kW)			
PV only	23.7	23.4	17.7
A: self-consumption	23.7	23.4	17.2
B: peak windows	17.2	21.3	21.1
C: grid cap	9.0	8.8	8.5

B. Resource, Seasonality and Battery Use

The reconstructed weather has 4% fewer cooling degree-days than the ASHRAE data for Sharjah [1]. PV produces 86.1 MWh a year (1,721 kWh/kWp). Summer PV output is only about 15% above winter output, because heat losses offset the longer days, while building demand almost doubles (Fig. 2). Storage therefore cannot remove the summer deficit.

Without auxiliaries, Rule A raises the share of demand met by PV from 50–61% to 81–82% (Table IV). The battery makes only 104–163 equivalent full cycles a year, not the 365 assumed for the year-10 capacity (Fig. 4). Most of the remaining import falls in May to October, and a larger daily battery cannot remove it.

C. Operating Rules and Useful Capacity

Fig. 5 shows a summer day. Under Rule A the battery charges from the midday PV surplus and is empty by about 20:30. Rule B covers its discharge windows but imports more over the year, because energy held overnight leaves less room for the next day's PV surplus (Table IV). Rule C cuts the highest import by more than half at little cost in energy, except in the office.

In the retail and 24/7 buildings 86–99 kWh usable captures 95% of the largest self-consumption benefit; the office needs about 195 kWh to carry PV across the weekend (Fig. 4). The year-10 capacity of 132 kWh still keeps most of the benefit. The spare capacity could serve as a backup reserve, which was not simulated.

D. Cooling, Efficiency and Technology

Fig. 6 shows the effect of the envelope in the 24/7 LFP case. On 15 July the bare container needs up to 11 kW of heat removal near midday. With 50 mm PU it needs about 1.5 kW, and this load follows the battery's charging and discharging

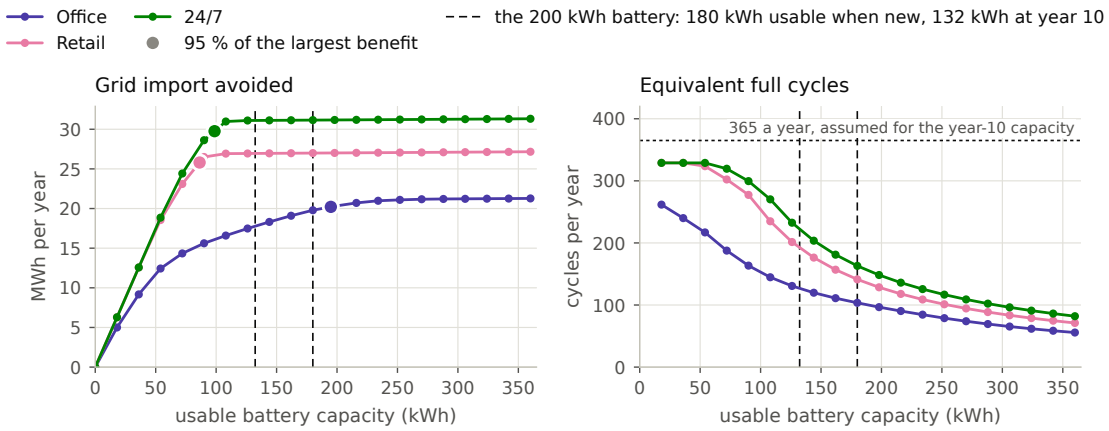


Fig. 4. Physical capacity sweep under Rule A at a fixed 50 kW, without storage auxiliaries: grid import avoided with the 95% benefit point (left) and equivalent full cycles (right).

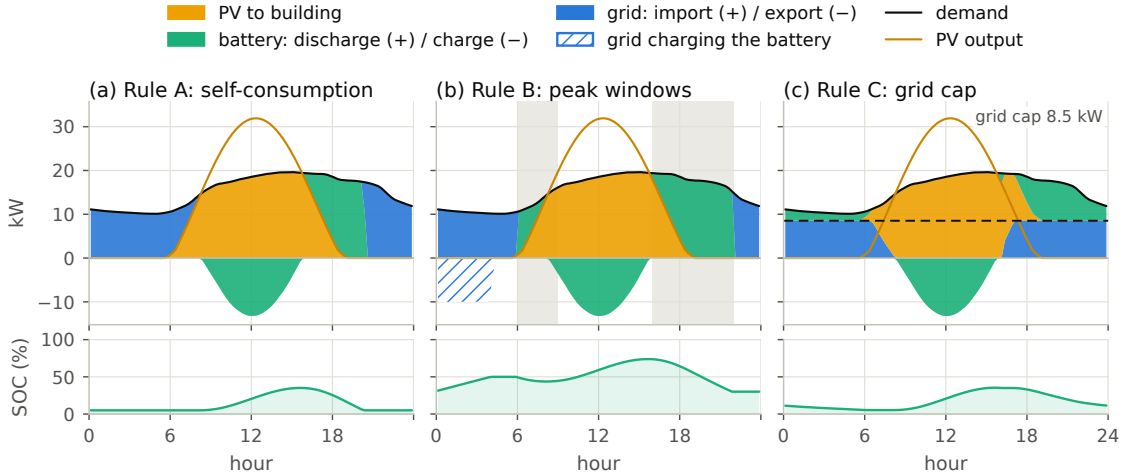


Fig. 5. One summer day (15 July), 24/7 building, without storage auxiliaries: power flows (top) and state of charge (bottom). Shaded: Rule B discharge windows.

rather than the sun. Over the year, bare steel needs five to six times as much cooling electricity as the insulated LFP container (Table V). White coating alone saves about 40%, insulation most of the rest.

Fig. 7 compares the two batteries. When own use is counted, insulated LFP delivers 77–81% of the energy it takes in and VRFB 61–64%. The office has the lowest efficiency because the constant auxiliaries are spread over the fewest cycles. The largest losses of LFP are its controls; those of VRFB are its stack and pumps, and the stack losses also add to its cooling (Table V). Insulating the container avoids about as much grid import as choosing LFP over VRFB: the envelope matters as much as the chemistry.

E. Economics and Break-Even Conditions

Year-one bill savings are only 1–4% of the installed cost, and simple payback is longer than the battery life in every case (Table VI). LFP loses less than VRFB over 20 years, despite its replacement in year 15. The LFP LCOS is within the 0.19–0.65 $/kWh found for an Abu Dhabi commercial building [10]; the VRFB LCOS is higher.

The lower part of Table VI shows how far each input must move. The break-even LFP price is below even the 2024 average for utility-scale storage, 192 $/kWh [24], and VRFB would need a sixth or less of the 360 $/kWh paid in Chinese tenders [25]. A smaller battery loses less, but no size from 50 to 300 kWh pays at today's prices. Only several favorable changes together make one case, 24/7 LFP, pay.

PV alone pays back at 1,000 $/kW [23], but adding either battery makes the joint project negative (Table VI), so the two investments should be judged separately.

F. Value Left, Paid Exports and the Better Buy

Value left at year 20 changes little (Table VII). LFP is unchanged, because its value left was already counted, and even selling the VRFB for half its initial cost leaves both batteries negative.

Paid evening exports change the dispatch. The optimizer saves PV for the evening and exports it, so the building imports

TABLE V
COUPLED ANNUAL RESULTS UNDER RULE A, 200 KWH AND 50 KW, INSULATED CONTAINER

	LFP			VRFB		
	Office	Retail	24/7	Office	Retail	24/7
Grid import avoided, net of own use (MWh/year)	17.0	23.7	27.9	13.7	18.0	21.4
Cooling electricity (MWh/year)	1.3	1.5	1.6	2.4	2.8	3.2
same, bare dark steel	7.7	7.8	7.9	8.4	8.7	9.0
Pump electricity (MWh/year)	0	0	0	1.2	1.6	1.8
Highest cooling heat load (kW)	1.9	1.7	1.5	4.7	4.5	3.8
Per 100 kWh taken in (kWh)						
lost in converter	4.1	4.2	4.3	3.5	3.6	3.7
lost in cells or stack	3.3	3.4	3.5	12.9	13.2	13.4
used for cooling	5.1	4.5	4.2	8.4	8.2	8.0
used by pumps	0	0	0	4.4	4.5	4.6
used by controls	10.4	8.0	7.0	9.3	7.6	6.6
delivered, i.e. total efficiency η_{tot} (%)	77.1	79.8	81.0	61.4	62.8	63.7

Insulated: white coating and 50 mm PU. Avoided import is relative to PV only. "Taken in" is the denominator of (18); self-discharge is at most 0.1 kWh per 100 kWh.

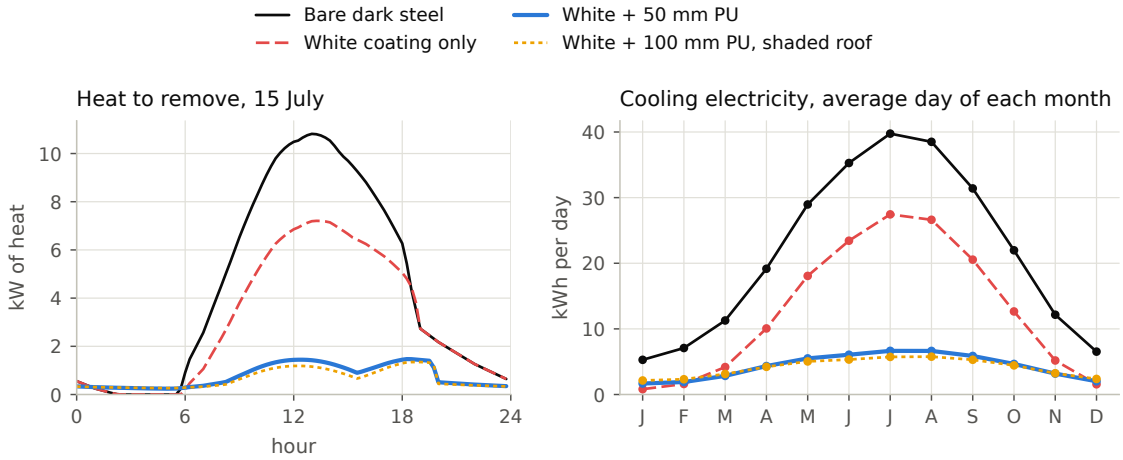


Fig. 6. Four envelopes, LFP, 24/7 building: heat to remove on 15 July and cooling electricity on the average day of each month. The thick line is the reference envelope. Annual-model results without a cooling-capacity limit, not a 45 °C sizing test.

more but the battery cycles more often, about 130 to 155 times a year over its full usable window. LFP pays back at an export price above about 0.54 $/kWh; VRFB would need about 0.97 $/kWh.

LFP is the better buy in every case but one. VRFB leads only when its future price, a 30-year life and resale of its electrolyte are assumed together, and even then both NPVs are negative (Table VII). Fig. 8 sums up the comparison for the retail building: LFP is ahead on six of the eight criteria, and VRFB only on life and cycle life.

IV. DISCUSSION

A. What Decides the Value of the Battery

Both chemistries reduce grid import, but the yearly saving is small against the capital cost, and with few cycles a year the fixed auxiliaries and O&M are spread over little energy. The long cycle life of the VRFB is not used, while its lower efficiency raises its cooling and lowers its useful energy, so LFP leads here, though not for every duration or market.

This agrees with studies that find PV profitable and storage rarely so under flat energy tariffs [10]–[12]: the tariff decides the value. The battery has value only because exports earn nothing; with net metering at the retail price, as in Dubai [26], neither battery would pay back at any price. An evening price signal, such as a time-of-use tariff [12] or the paid exports of Scenario 3, rewards storage even when grid import rises.

The envelope is the cheapest energy measure: for LFP, white coating and 50 mm PU save about 6 k$ of cooling electricity over 20 years. Its cost is already inside the system prices used, so this is the value of the measure, not its payback.

B. Thermal and Aging Interpretation

The annual model gives insulated LFP heat peaks about half of the 3.3 kW design value, because it uses the real orientation of each surface, the actual dispatch and a weather year that peaks at 42.9 °C; the design-day value, not the annual peak, sizes the air conditioner.

The annual model also checks the delivered-energy chain: the own use of insulated LFP, 13–18% of the charged energy,

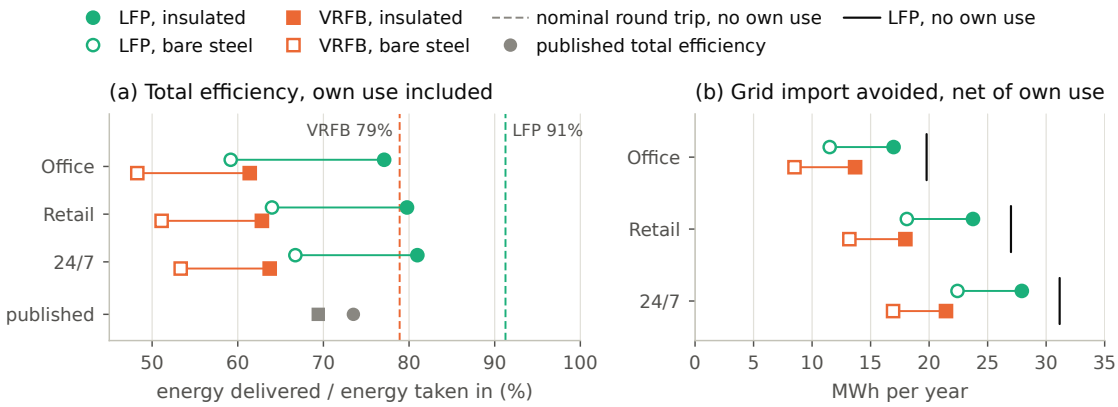


Fig. 7. LFP and VRFB in insulated and bare steel containers: (a) total efficiency η_{tot}; (b) grid import avoided. Published values: Li-ion and VRB sites [18].

TABLE VI
ECONOMICS OF INSULATED 200 KWH STORAGE AND WHAT IS NEEDED FOR NPV = 0

	LFP			VRFB		
	Office	Retail	24/7	Office	Retail	24/7
Year-one bill saving, before O&M (k$)	2.02	2.85	3.35	1.64	2.15	2.57
Simple payback (years)	56	36	30	495	157	101
NPV over 20 years (k$)	−78	−69	−64	−116	−111	−107
LCOS ($ per kWh delivered)	0.56	0.40	0.34	0.82	0.66	0.57
Needed for NPV = 0, one change at a time						
Battery price, 200 kWh ($/kWh)	63	106	128	12	37	57
Battery price, 100 kWh and 25 kW ($/kWh)	103	201	220	59	142	180
Electricity price ($/kWh)	0.62	0.42	0.35	0.98	0.75	0.63
Backup or other services (k$ per year)	8.0	7.0	6.5	11.8	11.3	10.9
Demand charge, battery capping the grid ($ per kW-month)[a]	54	44	67	79	70	112
Battery price for NPV = 0, several changes together ($/kWh)[b]	217	405	438	156	314	387
NPV of PV and battery together (k$)[c]	−56	−54	−55	−94	−96	−98

[a] Rule C runs without own use; SEWA has no demand charge. [b] 100 kWh, 5% discount rate, tariff +50%, LFP life 20 years. [c] PV at 1,000 $/kW [23], exports unpaid; PV alone: NPV +22/+15/+9 k$, payback 6.8–8.3 years. Year-one LFP values include capacity fade.

TABLE VII
SCENARIOS FOR THE RETAIL BUILDING

NPV over 20 years (k$)	LFP	VRFB
Today (Table VI)	−69.1	−110.7
Battery halved to 100 kWh and 25 kW	−24.2	−44.8
Electricity price doubled, 0.24 $/kWh	−41.1	−89.5
VRFB at its future price, 391 $/kWh [8]	−69.1	−70.8
VRFB life 30 years, electrolyte resold	−69.1	−95.6
VRFB future price, 30 years and resold	−69.1	−58.5
Scenario 2: value left for both	−69.1	−102.8
variant: LFP sold for 0, VRFB for $0.5C_0$	−75.1	−98.1
Scenario 3: evening exports at 0.30 $/kWh	−39.1	−81.6
at 0.60 $/kWh	+10.3	−45.2
export price for NPV = 0 ($/kWh)[a]	0.54	0.97

Scenario 3 keeps the Scenario 2 value left. Exports are a hypothetical contract, not a current utility offer. [a] NPV is linear between the two export prices; the VRFB value is extrapolated.

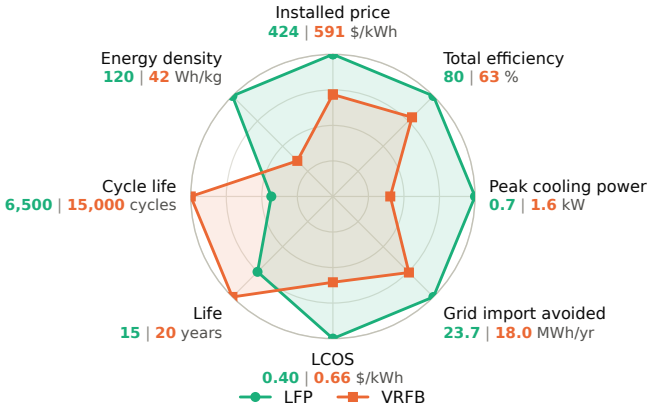


Fig. 8. LFP and VRFB on eight criteria, retail building, insulated container. Each axis is scaled so that the better battery scores 1; for price, peak cooling power and LCOS the lower value is better. Values under each name: LFP | VRFB. Cycle life and energy density are the middle of the ranges in [6].

is inside the range of [4], and the battery cycles far less than the 365 times a year assumed for the year-10 fade, so that estimate is cautious. The cell-to-air temperature offset is unknown and matters: at the +9.4 K of [2], about 357 kWh instead of 300 kWh would be needed for 200 kWh usable at year 10. Temperature, partial cycles and calendar aging should be modeled together before the replacement year is fixed [5], [27].

C. Limitations

The demand profiles are synthetic and the solar input is a smoothed average day, so cloud sequences, dust storms and weather extremes are absent; the lumped PV loss does not model cleaning or module aging [28].

Pump power, controls load, COP, self-discharge and the effective VRFB capacity need supplier data. VRFB layout, electrolyte containment and fit in the container are not checked. The model also ignores the thermal mass of the 8.3 m^3 VRFB electrolyte, which could be cooled at night and absorb heat during the day, so the VRFB cooling load may be overstated. Valuing every avoided kWh at the top slab favors storage in general. The export scenarios use ideal hourly forecasts and keep the base fade law although the battery cycles more.

V. Conclusion

For 50 kWp of PV and 100 MWh a year of synthetic demand in Sharjah, the container envelope, not the battery, sets the cooling: 83% of the 45 °C heat load of a bare steel container enters through its walls and roof, and white coating with 50 mm insulation cuts the load almost five-fold. Counting cooling, pumps and controls, insulated LFP delivers 77–81% of the energy taken in and VRFB 61–64%, and the envelope saves about as much grid energy as the choice of chemistry. The 200 kWh battery cycles only 104–163 times a year, and 86–99 kWh usable captures 95% of the self-consumption benefit in the retail and 24/7 buildings.

At 0.12 \$/kWh with exports unpaid, neither battery pays back over 20 years. LFP, with the smaller loss, needs an installed price of 63–128 \$/kWh, electricity at 0.35–0.62 \$/kWh, other services worth 6.5–8.0 k\$ a year or an evening export price above about 0.54 \$/kWh. VRFB is the better buy only with a future price, a 30-year life and resold electrolyte together. PV should therefore be decided on by itself, storage sized to measured demand and defined services such as backup, and the auxiliary energy and 45 °C cooling design checked against supplier data before purchase.

References

[1] ASHRAE, *ASHRAE Handbook: Fundamentals*. Atlanta, GA, USA: ASHRAE, 2017, SI ed., chs. 14 and 18.

[2] H. Shi, W. Xu, X. Zhu, J. Wang, K. Yang, Y. Zou, and Z. Chen, "A thermal-optimal design of lithium-ion battery for the container storage system," *Energy Science & Engineering*, vol. 10, no. 3, pp. 951–961, 2022.

[3] Y. Dong, Y. Ding, K. E. Elfeky, Y. Qi, W. Chu, and Q. Wang, "Thermal management for a stadium power supply container using a rack-level air cooling strategy," *Energies*, vol. 17, no. 7, p. 1654, 2024.

[4] W. Olis, D. Rosewater, T. Nguyen, and R. H. Byrne, "Impact of heating and cooling loads on battery energy storage system sizing in extreme cold climates," *Energy*, vol. 278, p. 127878, 2023.

[5] M. C. Yagci, O. Richter, R. Behmann, and W. G. Bessler, "Degradation modes of large-format stationary-storage LFP-based lithium-ion cells during calendaric and cyclic aging," *Journal of Energy Storage*, vol. 124, p. 116774, 2025.

[6] S. Coccato, K. Barhmi, I. Lampropoulos, S. Golroodbari, and W. van Sark, "A review of battery energy storage optimization in the built environment," *Batteries*, vol. 11, no. 5, p. 179, 2025.

[7] E. H. Kirk, F. Fenini, S. N. Oreiro, and A. Bentien, "Temperature-induced precipitation of V_2O_5 in vanadium flow batteries: Revisited," *Batteries*, vol. 7, no. 4, p. 87, 2021.

[8] N. Poli, C. Bonaldo, M. Moretto, and M. Guarnieri, "Techno-economic assessment of future vanadium flow batteries based on real device/market parameters," *Applied Energy*, vol. 362, p. 122954, 2024.

[9] M. He, D. Chartouni, D. Landmann, and S. Colombi, "Safety aspects of stationary battery energy storage systems," *Batteries*, vol. 10, no. 12, p. 418, 2024.

[10] N. Alkaabi, R. Jaradat, and A. Mayyas, "Evaluating the influence of operational uncertainty on solar photovoltaic system design and economic viability: Case study for commercial building in Abu Dhabi," *Energy Science & Engineering*, vol. 13, no. 12, pp. 6100–6113, 2025.

[11] N. G. Chatzigeorgiou, S. Theocharides, G. Makrides, and G. E. Georghiou, "A review on battery energy storage systems: Applications, developments, and research trends of hybrid installations in the end-user sector," *Journal of Energy Storage*, vol. 86, p. 111192, 2024.

[12] Y. Badiei and J. C. do Prado, "Analyzing the impact of electricity rates on the feasibility of solar PV and energy storage systems in commercial buildings: Financial vs. resilience perspective," *Energies*, vol. 16, no. 5, p. 2421, 2023.

[13] Dubai Electricity and Water Authority, "Standards for distributed renewable resources generators connected to the distribution network," DEWA, Dubai, UAE, Shams Dubai Version 2.0, Mar. 2016.

[14] Sharjah Electricity, Water and Gas Authority, "Energy calculator," Online, accessed: Oct. 19, 2026. [Online]. Available: https://sewa.gov.ae/en/energy-calculator

[15] J. A. Duffie and W. A. Beckman, *Solar Engineering of Thermal Processes*, 4th ed. Wiley, 2013.

[16] R. G. Allen, L. S. Pereira, D. Raes, and M. Smith, "Crop evapotranspiration: Guidelines for computing crop water requirements," FAO, Rome, Italy, Tech. Rep. Irrigation and Drainage Paper 56, 1998. [Online]. Available: https://www.fao.org/4/x0490e/x0490e00.htm

[17] A. Afshari, C. Nikolopoulou, and M. Martin, "Life-cycle analysis of building retrofits at the urban scale: A case study in United Arab Emirates," *Sustainability*, vol. 6, no. 1, pp. 453–473, 2014.

[18] L. da Silva Lima, M. Quartier, A. Buchmayr, D. Sanjuan-Delmás, H. Laget, D. Corbisier, J. Mertens, and J. Dewulf, "Life cycle assessment of lithium-ion batteries and vanadium redox flow batteries-based renewable energy storage systems," *Sustainable Energy Technologies and Assessments*, vol. 46, p. 101286, 2021.

[19] *ISO 5151:2017: Non-ducted air conditioners and heat pumps: Testing and rating for performance*, 3rd ed., International Organization for Standardization, Geneva, Switzerland, 2017.

[20] A. Chadly, R. R. Urs, M. Wei, M. Maalouf, and A. Mayyas, "Techno-economic assessment of energy storage systems in green buildings while considering demand uncertainty," *Energy and Buildings*, vol. 291, p. 113130, 2023.

[21] National Renewable Energy Laboratory, "Annual Technology Baseline 2024: Commercial battery storage," 2024, accessed: Oct. 19, 2026. [Online]. Available: https://atb.nrel.gov/electricity/2024/commercial_battery_storage

[22] International Renewable Energy Agency, "Electricity storage and renewables: Costs and markets to 2030," IRENA, Abu Dhabi, UAE, Tech. Rep., 2017.

[23] Y. Miky, A. G. Olabi, H. Rezk, E. T. Sayed, U. H. Issa, A. A. Shouny, and M. A. Abdelkareem, "Techno-economic feasibility and sensitivity analysis of solar photovoltaic/battery for DC power in commercial buildings (NEOM city)," *Thermal Science and Engineering Progress*, vol. 44, p. 102049, 2023.

[24] International Renewable Energy Agency, "Renewable power generation costs in 2024," IRENA, Abu Dhabi, UAE, Tech. Rep., 2025.

[25] Vanitec, "China sees surge in 100MWh vanadium flow battery energy storage projects," Online, Aug. 2024, accessed: Oct. 8, 2026. [Online]. Available: https://vanitec.org/latest-from-vanitec/article/china-sees-surge-in-100mwh-vanadium-flow-battery-energy-storage-projects

[26] M. Alsalman, V. Ahmed, Z. Bahroun, and S. Saboor, "An economic analysis of solar energy generation policies in the UAE," *Energies*, vol. 16, no. 7, p. 3165, 2023.

[27] N. Collath, B. Tepe, S. Englberger, A. Jossen, and H. Hesse, "Aging aware operation of lithium-ion battery energy storage systems: A review," *Journal of Energy Storage*, vol. 55, p. 105634, 2022.

[28] B. Adothu, S. Kumar, J. J. John, G. Oreski, G. Mathiak, B. Jäckel, V. Alberts, J. B. Jahangir, M. A. Alam, and R. Gottschalg, "Comprehensive review on performance, reliability, and roadmap of c-Si PV modules in desert climates: A proposal for improved testing standard," *Progress in Photovoltaics: Research and Applications*, vol. 32, pp. 495–527, 2024.